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EE9-CS7-30

IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS

APPLIED ADVANCED OPTIMISATION

Thursday, 8 May 2025, 10:00 AM

Time allowed: 3:00 hours

There are THREE questions on this paper.

Answer ALL questions.

All questions carry equal marks.

Any special instructions for invigilators and information for candidates are on page 1.

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Second Marker(s) : S.A. Evangelou

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APPLIED ADVANCED OPTIMISATION

Information for candidates:

- This paper has no additional information required.

1. a) In a smart city, a central charging station is responsible for charging N electric vehicles (EVs) overnight. Each EV i (with $i = 1,\ldots,N$) requires a total energy amount of at least E_i (in kWh) to be fully charged by the morning. Assume that the total charging period is discretized into D equal time slots of duration Δt. We define the non-negative charging power (in kW) allocated to each EV i in each time slot k as p_{ik}. The charging station has a maximum power capacity of $P_{\max}$ that cannot be exceeded at any moment. The electricity (energy) cost varies over time. This is captured by a cost-per-unit-of-time convex function $C(\cdot)$ of the total power allocated in a certain time slot.

Formulate the above scheduling problem to charge the EVs, i.e. allocating the power, as a convex optimisation problem in which you minimise the total cost over the charging period. [8 marks]

Hint: recall that dimensionally "energy" is equal to "power" times "time".

b) A busy four-way intersection in a metropolitan area experiences significant delays during peak hours. To alleviate congestion, the city's traffic management centre is tasked with optimizing the green signal times for the four approaches (labeled 1, 2, 3, and 4) in each fixed-length cycle of duration T seconds. For each approach i (with $i = 1,2,3,4$), let t_i (in seconds) be the green time allocated during each cycle. Due to different traffic patterns, each approach has a sensitivity parameter $a_i > 0$ that quantifies how the delay increases when the green time is insufficient. The delay d_i on approach i is modeled by a function which is the product of the sensitivity times the square of the non-green time for that approach. Safety regulations require that each approach receives at least a minimum green time, denoted by $t_i^{\min}$ for $i = 1,2,3,4$. In addition, the entire cycle time T must be allocated among the approaches. To promote fairness and avoid significant discrepancies in waiting times between approaches, the difference between the longest and shortest green times should not exceed δ seconds.

Formulate the above scenario as a convex optimization problem in which you want to minimize the total delay. [8 marks]

c) We have N points in $\mathbb{R}^2$ and a list of pairs of these points that must be connected by links. The positions of some of the N points are fixed; the placement and location problem consists in determining the positions of the remaining points, i.e., to place the remaining points. The objective is to place the points so that some measure of the total interconnection length of the links is minimized. The problem can be described in terms of an undirected graph with N nodes, representing the N points, see for instance Figure 1.1. With each node we associate a variable $x_i \in \mathbb{R}^2$ which represents its location or position. The problem is to minimize

$$\sum_{(i,j)\in\mathscr{A}} f_{ij}(x_i,x_j)$$

where $\mathscr{A}$ is the set of all links in the graph and f_{ij} is a cost function associated with arc (i,j). In particular, we consider problems where f_{ij} is a penalty function h of the distance associated with arc (i,j), namely $f_{ij}(x_i,x_j) := h(||x_i - x_j||_2)$.

We consider two instances of this problem, namely

i) $f_{ij}(x_i,x_j) = ||x_i - x_j||_2$, corresponding to the penalty function $h(z) = z$.

ii) $f_{ij}(x_i,x_j) = ||x_i - x_j||_2^2$, corresponding to the penalty function $h(z) = z^2$.

We solve these two versions of the problem and we obtain the placements in

Figure 1.1 where the 6 black dots indicate the free points, the 8 white squares indicate the fixed points, and 27 links are indicated with dotted lines.

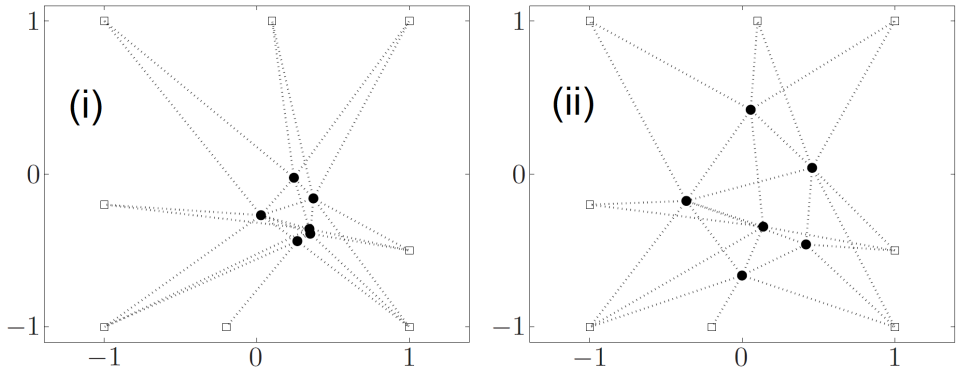


Figure 1.1 **Left:** Placement problem in instance (i). **Right:** Placement problem in instance (ii).

Compare the two placements in Figure 1.1 and explain the resulting distributions of the free points. Be as detailed as possible. [4 marks]

2. a) Consider the optimisation problem

$$\begin{aligned} \min \quad & x \\ s.t. \quad & (x-1)(x-2) \leq 0, \end{aligned}$$

with variable $x \in \mathbb{R}$.

i) Determine the feasible set, the optimal value, and the optimal solution. Plot the objective function and the constraint function versus x (in the same plot). In the plot, clearly indicate the feasible set and the optimal value.

[4 marks]

ii) Determine the Lagrange dual function g.
Hint: there are two cases to consider depending on the value of λ.

[4 marks]

iii) State the dual problem. Find the dual optimal value and the dual optimal solution λ^*. Does strong duality hold? [4 marks]

iv) Let $p^*(u)$ denote the optimal value of the perturbed problem

$$\begin{aligned} \min \quad & x \\ s.t. \quad & (x-1)(x-2) \leq u, \end{aligned}$$

as a function of the parameter u. Determine $p^*(u)$. Does $\frac{dp(u)}{du}|_{u=0} = -\lambda^*$ hold?
Hint: note that for $u < -\frac{1}{4}$ *the problem is infeasible. Thus, consider only the case* $u \geq -\frac{1}{4}$. [4 marks]

b) Consider the equality constrained least-squares problem

$$\begin{aligned} \min \quad & (Ax-b)^\top (Ax-b) \\ s.t. \quad & Gx = h, \end{aligned}$$

where $A \in \mathbb{R}^{m \times n}$ with rank $A = n$, and $G \in \mathbb{R}^{p \times n}$ with rank $G = p$.

Give the KKT conditions and determine the expression of the dual solution ν^*.
Hint: recall that $x^\top A^\top b = b^\top Ax$ *and that* $\frac{dx^\top A^\top Ax}{dx} = 2x^\top A^\top A$. [4 marks]

3. a) Sketch the following sets.

i) The convex hull of the set

$$\{(x_1,x_2) \in \mathbb{R}^2 : (0,0),(1,0),(2,0),(0,1),(1,1),(2,2)\}.$$

[2 marks]

ii) A set with minimal points (but no minimum) with respect to the cone $\mathbb{R}_- = -\mathbb{R}_+$. [2 marks]

b) Answer the following convexity questions.

i) Consider the function $f(x_1,x_2) = ax_1^2 + bx_1x_2 + cx_2^2 + dx_1 + ex_2 + f$, where a,b,c,d,e,f are constants. Find conditions on a,b,c,d,e,f to ensure that $f(x_1,x_2)$ is convex. [2 marks]

ii) Consider the function $f(x) = ae^{ax} + bx + c$, where a,b,c are constants. Find conditions on a,b,c to ensure that $f(x)$ is convex. [2 marks]

iii) Consider the function $f(x) = \frac{ax+b}{cx+d}$ in the domain $D = \{x : cx+d > 0\}$ where a,b,c,d are constants. Find conditions on a,b,c,d to ensure that $f(x)$ is convex in the domain D. [2 marks]

iv) Let $S \subset \mathbb{R}^2$ be defined as $S = \{(x_1,x_2) \in \mathbb{R}^2 : x_1^2 + x_2^2 \leq 1\}$. Prove the convexity of S using the definition of convex set.
Hint: note that $x_1y_1 + x_2y_2 \leq \sqrt{x_1^2+x_2^2}\sqrt{y_1^2+y_2^2}$. [2 marks]

v) Consider the hypograph of a convex function f defined as **hypo** $f = \{(x,t) : t \leq f(x)\}$. Prove that **hypo** f is a concave set using the definition of convex or concave set. [2 marks]

c) Formulate the problem

$$\begin{array}{ll} \min\limits_x & ||x||_1 \\ \text{s.t.} & ||Ax-b||_\infty \leq 1, \end{array} \tag{3.1}$$

where $A \in \mathbb{R}^{m\times n}$, $b \in \mathbb{R}^m$, and $x \in \mathbb{R}^n$, as a linear programme. Explain in detail the relation between problem (3.1) and the obtained linear programme.
[3 marks]

d) Consider the convex problem

$$\begin{array}{ll} \min & f_0(x) \\ s.t. & f_i(x) \leq 0, \quad i = 1,\ldots,m, \\ & Ax = b, \end{array}$$

with central path $x^*(t)$ for $t > 0$, defined as the solution of

$$\begin{array}{ll} \min & t f_0(x) - \sum_{i=1}^m \log(-f_i(x)) \\ s.t. & Ax = b. \end{array}$$

In this question we explore another parametrization of the central path. For $u > p^*$, let $z^*(u)$ be the solution of

$$\begin{array}{ll} \min & -\log(u - f_0(x)) - \sum_{i=1}^m \log(-f_i(x)) \\ s.t. & Ax = b. \end{array}$$

Show that the curve defined by $z^*(u)$, for $u > p^*$, is the central path. In other words, for each $u > p^*$, find $t > 0$ such that $x^*(t) = z^*(u)$, and conversely, for each $t > 0$, find $u > p^*$ for which $z^*(u) = x^*(t)$. [3 marks]